



iPhone 7 costs \$220

In case you were wondering how much does the iPhone 7 cost to manufacture.
<http://dgit.in/iPh7Cost>



Canon's M5 cam

The M5 mirrorless camera is Canon's best mirrorless camera, yet falls behind.
<http://dgit.in/CanonM5>

While your data science capabilities are important, it is equally important here that you should be able to deal with production code, analyse numbers, visualise information etc.

The interesting thing is, all of these companies will post their job requirements as 'Data Scientist', even though they require different skill sets and experience levels. Depending on the company you are applying for, you might want to give more importance to certain skills compared to others.

Skills needed

With the different requirements under the broad definition of 'data scientist', it's admittedly confusing whether one should focus on one particular skill, or another. But broadly, there are eight different areas where you need to focus:

- **Tools** - Be it a data centric company or a consulting company, you will definitely be expected to know the popular software tools for data science, namely statistical programming languages like R and SAS, along with a database query language like SQL.
- **Statistics** - This is where you go back to your basic stats class. Depending on the role, you may need up to an advanced level of knowledge of statistics for your job. It will definitely be a recurring topic in your interview, so it is better to be prepared. This is especially important for a role in a non-data company that is data driven, where you will be expected to present results, helping in decision making.
- **Multi-variable calculus and linear algebra** - If you're wondering why a data scientist would need to learn this, even though there are tools out there, programmed in R or other languages, that can take care of this, then you're not alone. The advantage here is that you might face questions based on these in your interview, since these form the basis for a lot of techniques used in data science. Also, if the company decides to build their own implementation in house, this will be needed.

- **Data Visualisation** - Often, people who will be using the results of your work will not have a technical in-depth knowledge of data science. Hence, it is really important to have visualisation skills which will help you communicate your data in an understandable form with the decision makers. Especially useful if you're in a company that is new to the data science field.
- **Software Engineering** - For smaller companies that are implementing data analysis and are hiring a data scientist for the first time, you can expect to be responsible for developing data driven products down the line and immediately take charge of other developments. A background in software engineering will be highly useful here.
- **Machine Learning** - In companies dealing with a large and complex data set, or where the product itself is data driven, a knowledge of

machine learning is a big plus. But it isn't exactly a deal breaker. Rather than the knowledge of programming an implementation, the knowledge of when to use which technique is more important here.

If you are an experienced professional, having worked on related fields earlier, it may or may not work in your favour. As Mr. Arun puts it, "Anyone who has worked with data, in analysing, parsing and interpreting, can hope to get started. However, to become a data scientist, it does need additional IT skills or at least an overall awareness and knowledge of technologies which might be difficult to catch up on for very experienced people".

Scope

It's good to know that everybody is going the 'data' way and there is a genuinely high demand for professionals in that area. But before you jump into the bandwagon and start preparing yourself, you need to know what kind of company you want to work for, based on the information given earlier. Why is this important? With this one decision, you might entirely change the kind of work you do.

E-commerce is a segment where data scientists and data experts are in high demand. The work there involves logistics optimisation, customer preference analysis, monitoring campaigns, showcasing sales performance to vendors and more. Another segment where data plays a big role are live apps – apps that connect service providers to consumers live – like a cab aggregation service, or a food ordering service. Such services usually have algorithms that are meant to optimise the service availability versus profitability in providing the service, which in turn is created on the basis of data!

Even the more traditional field of advertising has been leveraging data since the popularity of mobile advertising has grown immensely. Don't take our word for it. Mr. Sunil Nandihalli, Head of Data Sciences at AppLift, a mobile ad tech company, explains, "Data can be used in many ways to



MR. ARUN KRISHNAN
 FOUNDER AND CEO, NFACTORIAL
 ANALYTICAL SCIENCES

“Automation is a tool like any other. Data scientists would still be required to bring everything together into a coherent whole and in enabling interpretation and insights from the same data.”